Dropout risk in higher education via multidimensional features and machine learning

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Abstract

Student turnover is a major challenge for higher education institutions. Early identification of high dropout risk students is of great significance for improving retention rate and optimizing resource allocation. Based on multi-source educational data, this study constructs a machine learning prediction framework, aiming to accurately classify students into two categories: ' dropout ' and ' non-dropout '. By mining data sets containing multi-dimensional features such as demographic characteristics, socioeconomic status, and academic performance, various algorithms such as random forest, XGBoost, support vector machine, logistic regression, and stacked ensemble are compared. The results show that the ensemble learning method performs better in this prediction task. In addition, SHAP technology is used to analyze the attribution of model output. The experimental results show that the proposed model has advantages in F1 score and recall rate. The analysis of feature importance further reveals the core factors affecting students ' dropout, such as course pass rate and tuition payment status. The above findings provide highly interpretable decision support for education administrators, thereby enhancing the credibility and transparency of artificial intelligence in educational applications.

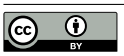
CCS Concepts

• **Computing methodologies**; • **Machine learning**; • **Machine learning approaches**; • **Ensemble methods**; • **Classification and regression trees**; • **Applied computing**; • **Education**; • **E-learning**;

Keywords

Student Dropout Prediction, Educational Data Mining, Machine Learning, Explainable Artificial Intelligence, SHAP Values, Educational AI Governance

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1 Introduction

With the ongoing expansion of higher education, the student dropout rate has become a key indicator for assessing both educational quality and the operational performance of colleges and universities. A high dropout rate not only causes a waste of educational resources, but also hinders the personal career development of students. Therefore, using data-driven methods to predict students' potential dropout behavior in advance has become a research hotspot in the field of Educational Data Science (EDS).

Traditional education evaluation mostly relies on post-hoc statistical data, which lacks real-time performance and individual pertinence. In recent years, the rapid development of machine learning technology has laid a solid foundation for building high-precision prediction models. Breakthroughs in deep learning and feature representation have shown the great potential of AI in processing complex data, which promotes the application of these technologies in education data analysis [1]. However, the application of advanced algorithms in the education field also faces unique challenges. With the increasing complexity of algorithms such as deep network and ensemble learning, the "black box" problem of the model is becoming more and more prominent [2]. This kind of opacity is inconsistent with the high requirements of fairness, transparency and accountability in the education field.

Under the framework of educational AI governance, algorithm decisions need to be understandable, so that educators can take targeted intervention measures. Based on the education data set with rich feature dimensions, this paper constructs a binary classification prediction model to distinguish dropout students from non-dropout students. Although advanced AI systems have achieved excellent results in many fields [3] the core contribution of this paper is to apply high-precision prediction ability to the scenario of student retention, and solve the bottleneck problem of model interpretability.

Specifically, this study has made three main contributions: Firstly, a data-driven binary classification framework is built, and the SMOTE technology is used to solve the problem of data imbalance. Secondly, the performance differences between single models and ensemble models are systematically compared. We draw upon advanced algorithmic strategies [4] to determine the optimal modeling method suitable for education data. Thirdly, SHAP value is used to analyze the interpretability and transparency of the model, revealing the internal decision logic of the "black box" model. This analysis conforms to the ethical requirements of educational AI governance transparency, and provides a reliable basis for human-machine collaborative education intervention.

2 Data and preprocessing

2.1 Dataset overview

The data set used in this study contains complete information of college students, and the data features cover multiple dimensions: Demographic features: marital status, gender, age at enrollment, nationality, etc. Socioeconomic features: parents' occupation and education level, study relocation status, scholarship acquisition, tuition arrears (Debtor, Tuition fees up to date), etc. Macroeconomic indicators: unemployment rate, inflation rate, gross domestic product (GDP) during enrollment, etc. Academic performance features: course unit information of the first and second semesters, including the number of registered courses, assessment times, passed courses and scores.

2.2 Data cleaning and label conversion

The target variables in the original data are divided into three categories: "Dropout", "Graduate" and "Enrolled". In order to focus on the prediction of dropout risk, this paper converts the research task into a binary classification problem: mark "Dropout" as 1 (positive sample), and merge "Graduate" and "Enrolled" into 0 (negative sample, i.e. non-dropout). The data set has a serious class imbalance problem, and non-dropout students account for the vast majority. If not processed, the model will tend to predict the majority class, resulting in a decrease in the recognition rate of dropout students.

2.3 Feature engineering and preprocessing

In order to adapt to the input requirements of machine learning algorithms, this paper constructs a standardized data preprocessing process [5]: For numerical features such as age, scores and macroeconomic indicators, the median is used to fill in missing values (SimpleImputer with strategy='median'), and then standardization (StandardScaler) is carried out to eliminate the influence of dimension differences on the model, especially for Support Vector Machine (SVM) and Logistic Regression (LR). For categorical features such as marital status, courses and application methods, the mode is used to fill in missing values, and one-hot encoding (OneHotEncoder) is used to convert categorical variables into numerical vectors to capture the independent relationship between different categories.

2.4 Correlation analysis

Before feature selection, this paper analyzes the correlation between numerical features and binary target variables (Target_Binary). The correlation heat map shows that some academic features are significantly negatively correlated with dropout status, such as the number of passed courses in the second semester; while tuition arrears and other features are positively correlated with dropout risk. This provides a preliminary statistical basis for the subsequent model interpretation.

2.5 Dataset splitting and resampling

In order to evaluate the generalization ability of the model, the data set is divided into training set (80%) and test set (20%) by stratified sampling, so as to keep the proportion of positive and negative samples in the training set and test set consistent [6]. Aiming at the class imbalance problem in the training set, this paper adopts the Synthetic Minority Oversampling Technique (SMOTE). This technology generates new synthetic samples by interpolating minority samples, balances the proportion of positive and negative samples, and effectively improves the model's sensitivity to dropout samples.

3 Model construction

This paper selects five representative machine learning algorithms for modeling, covering linear models, tree models and ensemble models, to find the optimal solution: Logistic Regression (LR): as the benchmark model, used to evaluate the linear separability of data. Support Vector Machine (SVM): using radial basis function (RBF) kernel function to process nonlinear relations. Random Forest (RF): an ensemble model based on Bagging strategy, with good anti-overfitting ability [7]. XGBoost: a gradient boosting tree model based on Boosting strategy, which usually has excellent performance on structured data. Stacking Ensemble: take Random Forest and XGBoost as base learners, and Logistic Regression as meta-learner (Final Estimator), integrate the advantages of different models to further improve prediction performance. All models are trained on the training set balanced by SMOTE, and tested on the test set with original data distribution..

4 Experimental results and analysis

4.1 Model performance evaluation

This paper comprehensively evaluates the model performance with Accuracy, Recall and F1 Score. In the scenario of dropout prediction, Recall is particularly important, because the cost of missing potential dropout students (False Negative) is higher than that of misjudging normal students (False Positive). F1 Score is the harmonic average of Precision and Recall, which is used to measure the comprehensive performance of the model. The performance of each model on the test set is shown below [8].

The experimental results show that there are obvious differences in the performance of each model[9]: Logistic Regression has high computational efficiency, but its ability to process complex nonlinear relations is weak; Random Forest and XGBoost have strong feature extraction ability, and their F1 scores and Recall rates are significantly higher than those of linear models; Stacking Ensemble model has a slight improvement in individual indicators, but also brings higher computational complexity. Considering the model performance and computational efficiency, XGBoost is selected as the optimal model for further in-depth analysis.

4.2 Confusion matrix analysis

The confusion matrix of the optimal model shows that the model has extremely high accuracy in predicting the "Non-Dropout" (0) category, and also has a high number of correct classifications for the "Dropout" (1) category. However, there are still a small number of False Negatives, that is, some dropout students are not identified by the model. This suggests that in practical application, the threshold adjustment strategy can be combined to further sacrifice part of the Precision in exchange for a higher Recall rate.

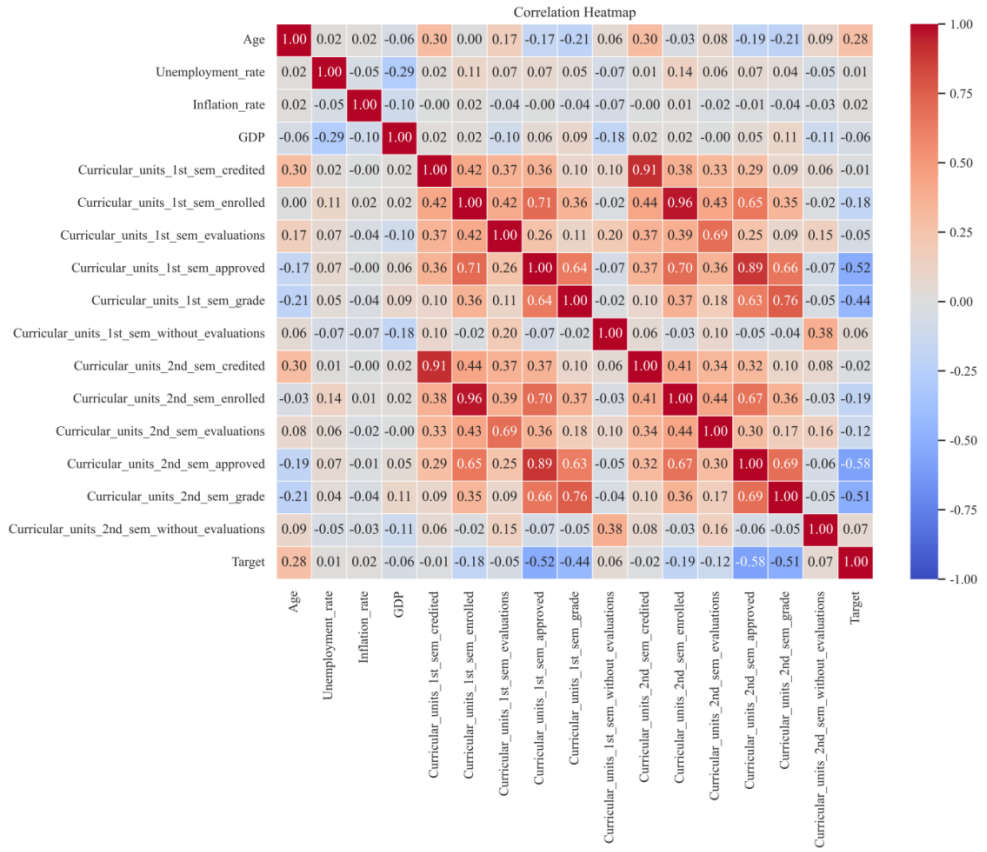


Figure 1: Correlation Heatmap

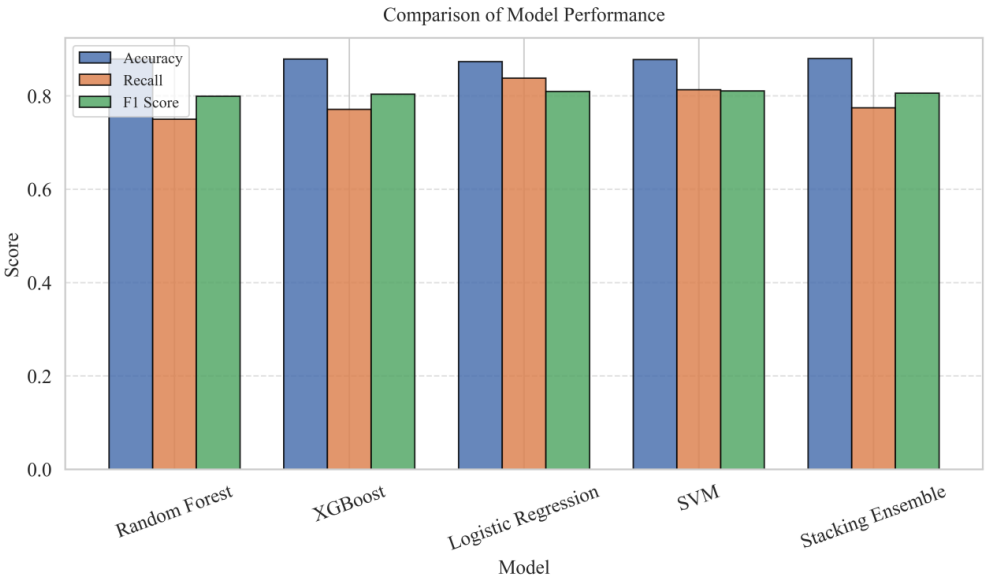


Figure 2: Comparison of Model Performance on the Test Set (Bar Chart)

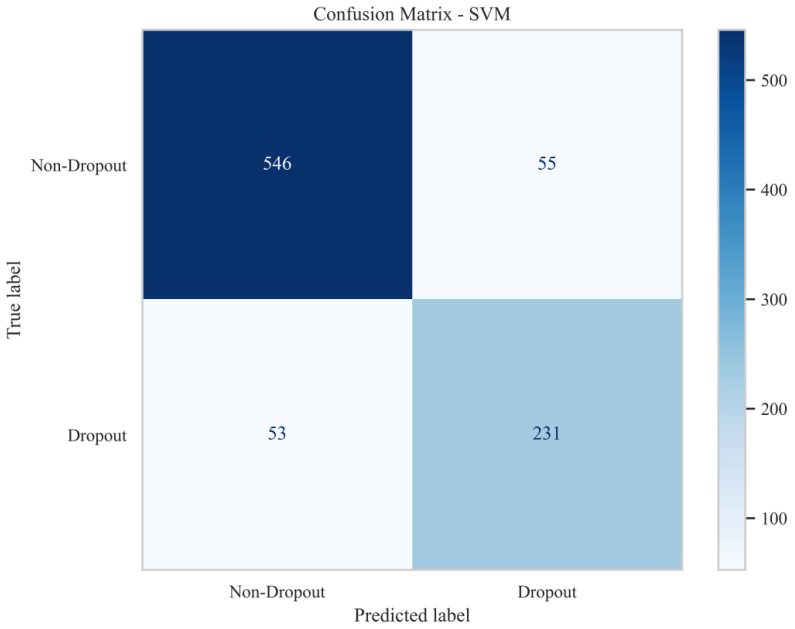


Figure 3: Confusion Matrix of the Optimal Model

5 Model interpretability and educational AI transparency

5.1 Global feature importance analysis: identifying key drivers of dropout

From a macro perspective, to clarify the main factors affecting students' dropout, this paper calculates the average absolute SHAP value of each feature in all test samples and sorts them. Figure 4 shows the top features contributing the most to the model prediction [10].

The analysis reveals the three main factors that lead to students' dropout and their practical significance. First of all, academic performance has been proved to be the most important predictor, among which the importance of the number of course recognitions and final scores obtained in the second semester is in the forefront. This shows that students ' real-time academic status can provide the most direct and reliable signal of dropout risk identification. Secondly, economic pressure also has a significant impact - tuition payment status as a key feature, reflecting economic difficulties or a weakening willingness to invest in higher education, often becomes an early warning signal for students to drop out early. In addition, enrollment-related characteristics have a continuous impact, especially the age of enrollment. Non-traditional students (such as older learners) usually face greater challenges in balancing work, family and academic responsibilities, and thus have a higher risk of dropout. In summary, this global feature importance analysis enables education managers to clarify the priorities of resource allocation, so as to monitor and intervene in student groups with declining academic performance or high arrears.

5.2 Directionality and distribution of feature impact

The global feature importance analysis can only reveal the overall impact of each feature on the prediction results, while the SHAP summary diagram further illustrates how each feature specifically affects the prediction results. The results show that the less number of courses passed in the second semester is the strongest driver of the high dropout risk. On the contrary, the more number of courses passed significantly reduces the risk. Tuition arrears are identified as strong high-risk signals, while timely payment helps to reduce risks. Older age at school is associated with a higher possibility of dropout, which may be due to the fact that this group bears more social responsibilities and is more difficult to integrate into campus life. In addition, obtaining scholarships has played a protective role, effectively reducing the risk of dropping out of school by providing economic support and ability recognition. In the context of educational artificial intelligence governance, SHAP analysis meets the key requirements of transparency and interpretability. It explains the risk drivers in a way that is consistent with the intuition of human experts, thereby enhancing trust in algorithmic decision-making and providing a basis for the implementation of precise interventions : academic counseling for curriculum-related risks, economic support for tuition problems, and psychological counseling for adaptation difficulties. Such targeted measures are more cost-effective than the ' one-size-fits-all ' approach.

6 Discussion and conclusion

This study uses a public data set to construct a binary classification prediction framework for students ' dropout behavior. After data cleaning, SMOTE oversampling processing and multi-model comparison and evaluation, a machine learning model with high

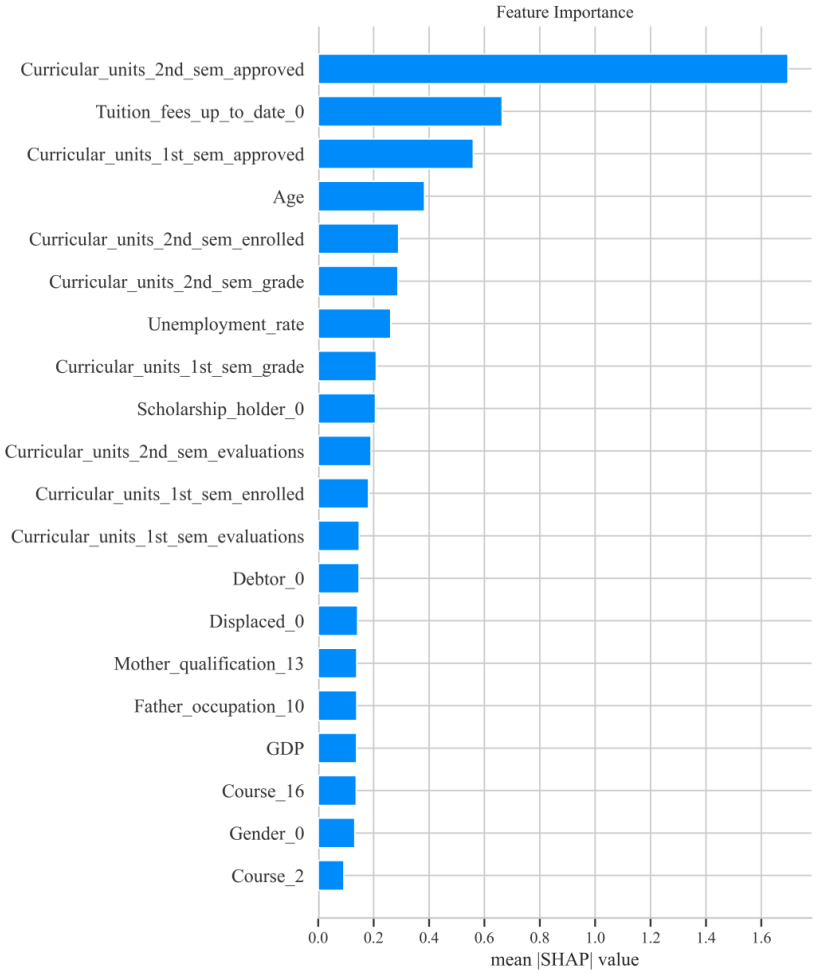


Figure 4: Feature Importance Ranking Based on Average SHAP Values (Bar Chart)

F1 score and recall rate was successfully trained. The typical representatives are XGBoost and random forest models. In addition, with the help of SHAP value, the interpretability analysis of the model is carried out. The study further confirms that academic performance and economic status are the key factors leading to students ' dropout, thus providing a practical reference example for the transparent governance of educational artificial intelligence system.

It should be pointed out that the prediction model faces important ethical responsibilities in the process of landing in the field of education and must be carefully dealt with. First, privacy and data governance : the use of socio-economic and demographic data may pose privacy risks. Future deployment should strictly abide by data protection laws and regulations, and actively adopt privacy protection technology to ensure that student data is only used for supportive purposes. Second, algorithm bias : machine learning models may unintentionally inherit historical bias from training data, which in turn causes injustice to certain groups. It is necessary to regularly review equity indicators to prevent models from exacerbating inequality in education. Third, non-discriminatory interventions: The core challenge is to transform the ' high-risk ' prediction results into specific action plans while avoiding labeling or stigmatizing students. The predicted results should not be used to negatively judge students, but should trigger a ' resource recommendation ' mechanism, such as pushing economic aid information or academic counseling recommendations.

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References

[1] Yann LeCun, Yoshua Bengio, Geoffrey Hinton. 2015. Deep Learning. Nature 521, 436–444. https://doi.org/10.1038/nature14539

[2] Kaiming He, Xiangyu Zhang, Shaoqing Ren, Jian Sun. 2016. Deep Residual Learning for Image Recognition. In 2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR). IEEE, 770-778. https://doi.org/10.1109/CVPR.2016.90

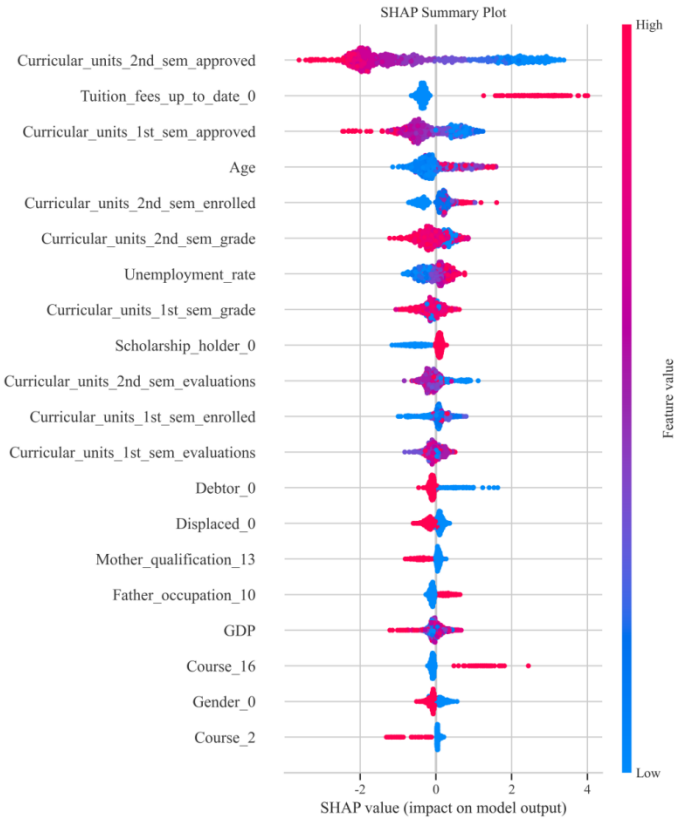


Figure 5: SHAP Summary Plot (Dot Plot) Showing the Positive and Negative Impacts of Feature Value Levels on Prediction Results (Dropout Probability)

[3] David Silver, Aja Huang, Chris J. Maddison, Arthur Guez, Laurent Sifre, George van den Driessche, Julian Schrittwieser, Ioannis Antonoglou, Veda Panneershelvam, Marc Lanctot, *et al.* 2016. Mastering the Game of Go with Deep Neural Networks and Tree Search. Nature 529, 484–489. https://doi.org/10.1038/nature16961

[4] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, Illia Polosukhin. 2017. Attention Is All You Need. In Advances in Neural Information Processing Systems 30 (NIPS 2017). 5998-6008. https://doi.org/10.48550/arXiv.1706.03762

[5] Alex Krizhevsky, Ilya Sutskever, Geoffrey E. Hinton. 2017. ImageNet Classification with Deep Convolutional Neural Networks. Communications of the ACM 60, 84–90. https://doi.org/10.1145/3065386

[6] Tianqi Chen, Carlos Guestrin. 2016. XGBoost: A Scalable Tree Boosting System. In Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining. 785–794. https://doi.org/10.1145/2939672.2939785

[7] Leo Breiman. 2001. Random Forests. Machine Learning 45, 5–32. https://doi.org/10.1023/A:1010933404324

[8] Sepp Hochreiter, Jürgen Schmidhuber. 1997. Long Short-Term Memory. Neural Computation 9(8), 1735-1780. https://doi.org/10.1162/neco.1997.9.8.1735

[9] Ian Goodfellow, Jean Pouget-Abadie, Mehdi Mirza, Bing Xu, David Warde-Farley, Sherjil Ozair, Aaron Courville, Yoshua Bengio. 2014. Generative Adversarial Nets. In Advances in Neural Information Processing Systems 27 (NIPS 2014). 2672-2680. https://doi.org/10.48550/arXiv.1406.2661

[10] David E. Rumelhart, Geoffrey E. Hinton, Ronald J. Williams. 1986. Learning Representations by Back-Propagating Errors. Nature 323, 533–536.